

TUMOR LOCALIZATION IN BREAST PHANTOMS

CNN vs MLP

A matched, head-to-head comparison of two machine-learning approaches for microwave tumor localization

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Preprocessing pipeline v2 (session-mean subtraction removed)

THE PROBLEM

Find a hidden tumor inside a breast phantom

- Goal: locate a small tumor surrogate (metal fishing weight) placed inside a realistic breast phantom, without opening it up.
- Microwave imaging: low-cost, non-ionizing alternative screening modality.
- The tumor perturbs how microwaves reflect and transmit through the tissue, but the change is tiny and buried in drift and noise.
- Worst case: a glandular insert (dense tissue) further masks the tumor signal.
- **Task: given one measurement, which of ~33–51 grid positions holds the tumor?**

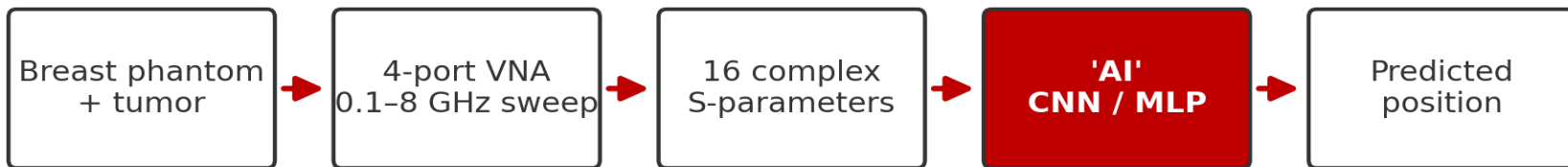


A3 breast phantom with F5 glandular insert and measurement grid

THE SOLUTION

Let a machine learn the fingerprint of every position

Classic inverse-scattering reconstruction is slow and fragile. Instead, we treat localization as a pattern-recognition problem: measure the phantom with the tumor at every known position, then train a model to recognise each position's electromagnetic fingerprint.



16

S-parameters per sweep

791

frequency points, 0.1–8 GHz

~2–3%

chance-level accuracy



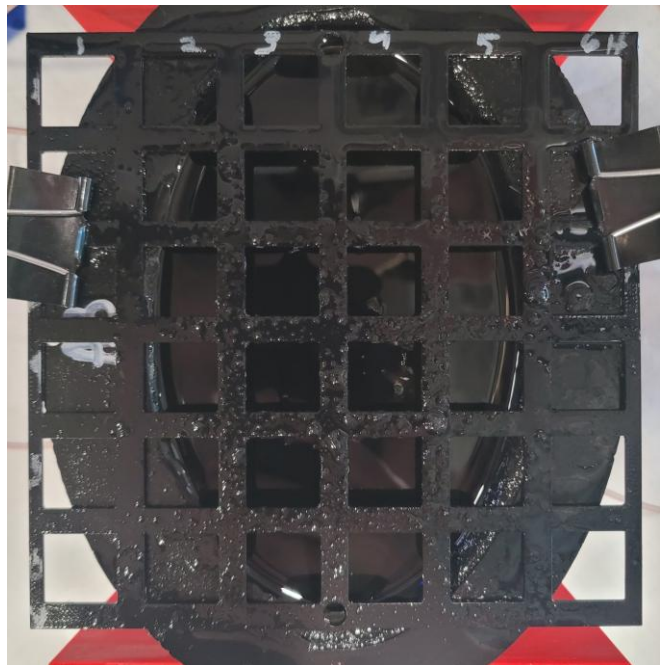
PART 1

Methods

The measurement, the two models, and how they differ

METHODS

The physical measurement



Empty A3 phantom: adipose bowl, 6×6 in grid, 4 antennas at the rim

- 4-port Hunter VNA sweeps 0.1–8 GHz (791 points) through 4 antennas around the phantom.
- Each sweep captures all 16 S-parameters: 4 reflections ($S_{11}...S_{44}$) plus transmissions between every antenna pair.
- Tumor surrogate is moved through a grid of cell corner positions; 16 repeat trials per position.
- Baseline (empty-phantom) sweeps recorded each session for calibration.
- **Each setup measured in 3–4 independent sessions on different setups/times; this matters later (LOSO).**

Two ways to turn a measurement into a position

Both models receive the SAME preprocessed measurement and output a probability for every grid position. They differ in how they look at the data.

CNN

Convolutional Neural Network · MATLAB

- Sees the measurement as a 2-D image: y axis = channels, x axis = frequency
- Convolution filters slide along frequency, learning local patterns

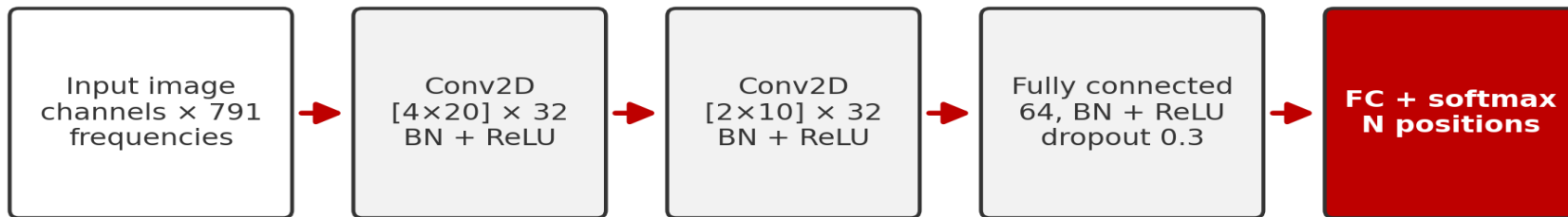
MLP

Multi-Layer Perceptron · Python

- Sees the measurement as one long flattened vector of numbers
- Every input connects to every neuron: the network has no built-in idea that neighboring frequencies are related, so any such structure must be learned from scratch

METHODS

The CNN



trained with Adam · 100 epochs · batch 16 · MATLAB Deep Learning Toolbox (GPU)

- Input is an image: one row per data channel (e.g. $|S_{11}|$, $\angle S_{11}$, ...), one column per frequency point (791).
- Two convolution layers (32 filters each) detect local patterns such as resonance shifts and ripples anywhere along the spectrum, sharing weights across frequency.
- Batch-normalisation + ReLU after every layer; 30% dropout fights overfitting.
- One network, trained 100 epochs with Adam (batch 16) on the GPU; softmax outputs one probability per grid position.
- Cost: 5-45 min per LOSO fold on a laptop GPU (input-size dependent); the slowest configs take hours per evaluation.

METHODS

The MLP



trained with Adam · early stopping on validation split · Python scikit-learn (CPU)

- Input is a flat vector: all channels and frequencies concatenated (thousands of features). No spatial structure is assumed.
- Two dense layers (256 $\rightarrow$ 128, ReLU): every feature can interact with every other, but nothing is shared or localized.
- Early stopping on a held-out validation split prevents overtraining.
- Ensemble: 3 networks with different random seeds are averaged, plus a logistic-regression model; the ensemble vote is the prediction.
- Cost: seconds to a few minutes per LOSO fold on a plain CPU; the full study sweep trains $\sim 10\times$ faster than the

METHODS

CNN vs MLP: structural differences

	CNN	MLP
How it sees data	2-D image (channels × frequency)	1-D flattened vector
Inductive bias	Local patterns, shared across frequency	None; fully general
Parameters	Small conv filters + FC (~0.5–2 M)	Huge first layer (features × 256)
Training	Fixed 100 epochs, single network	Early stopping, 3-seed ensemble + LogReg
Framework	MATLAB Deep Learning Toolbox (GPU)	Python scikit-learn (CPU)
Training time	5-45 min per LOSO fold (GPU); ~20 h for the full sweep	10 s-7 min per fold (CPU); ~3-4 h for the same sweep

Same job, opposite philosophies: the CNN assumes frequency-local structure; the MLP assumes nothing. The CNN buys its accuracy with compute: roughly 10x the training time, even with a GPU. Inference is milliseconds for both.

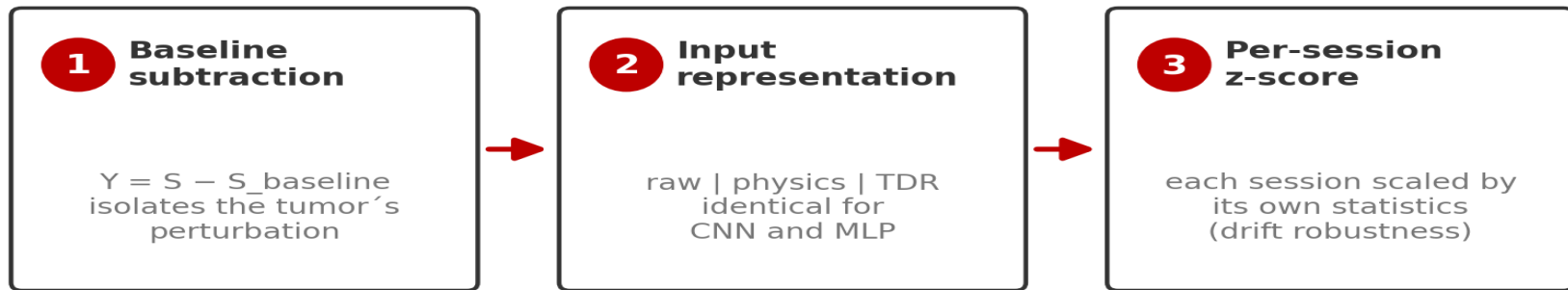


PART 2

The data pipeline

What happens to a measurement before any 'AI' ever sees it. Identical for both models.

Three steps from raw sweep to model input



v2: the session-mean subtraction step was removed (redundant with the per-session z-score)

- Step 1 removes everything that is not the tumor; step 3 (per-session z-score) is the key drift defense: each session is standardized by its own statistics. Applied identically to CNN and MLP.
- **v2 change: the per-session mean subtraction used in v1 was removed. An ablation showed it was redundant with the z-score for the MLP (up to +6 pts recovered on hard configs) and roughly neutral for the CNN outside TDR inputs.**

Three input representations

RAW

Magnitude + phase of each calibrated S-parameter.

The measurement as the VNA reports it. Used for the main results.

PHYSICS

Adds real & imaginary parts and the IFFT time-domain (TDR) envelope.

Derived at no cost from the same data; covered at the end.

TDR-ONLY

Just the time-domain envelope, a pulse-echo view.

An ablation: how much lives in time domain alone?

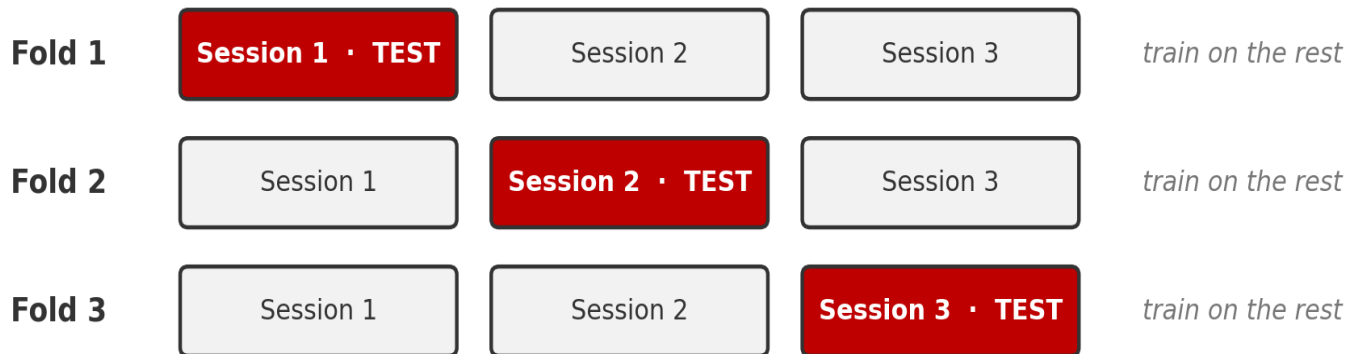
Same numbers, two shapes:

CNN: channels stacked as image rows $\times$ 791 frequency columns

MLP: the same rows laid end-to-end as one long vector

Antenna subsets work the same way: '2 antennas' keeps only S-parameters among ports 1 & 3; '1 antenna' keeps only S11.

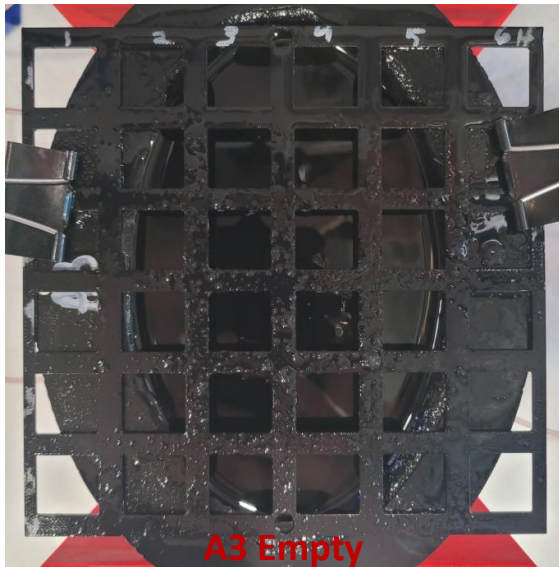
How we score: Leave-One-Session-Out



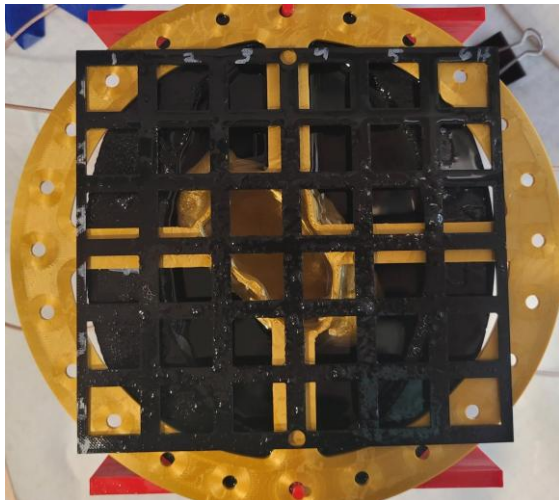
Leave-One-Session-Out: the model NEVER sees the test session, so the score reflects a brand-new day of measurements

- A random 75/25 split mixes trials from the same session into train AND test, so the model can cheat off session drift, inflating accuracy.
- LOSO trains on $N-1$ sessions and tests on the session it has never seen; every session takes a turn. We report per-position majority vote over its 16 trials.
- **We have thoroughly stress-tested LOSO and verified that it works: under LOSO, models cannot exploit session drift, and drift-only information scores near chance.**

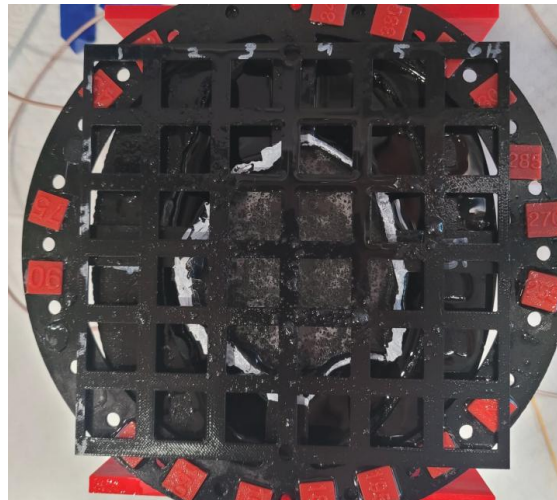
Three phantom setups, increasing difficulty



adipose only
51 positions · 3 sessions



small glandular insert
39 positions · 4 sessions



large glandular insert
33 positions · 3 sessions (latest)



PART 3

Results

Per-position vote accuracy (%) under LOSO · identical raw inputs (mag + phase) · chance = 2–3% · pipeline v2

All 4 antennas: full S-parameters (16)

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	99.3 ± 1.1	99.3 ± 0.9	+0.0
A3 + F4	39-way	97.4 ± 3.0	98.7 ± 2.2	-1.3
A3 + F5	33-way	100.0 ± 0.0	94.9 ± 3.8	+5.1

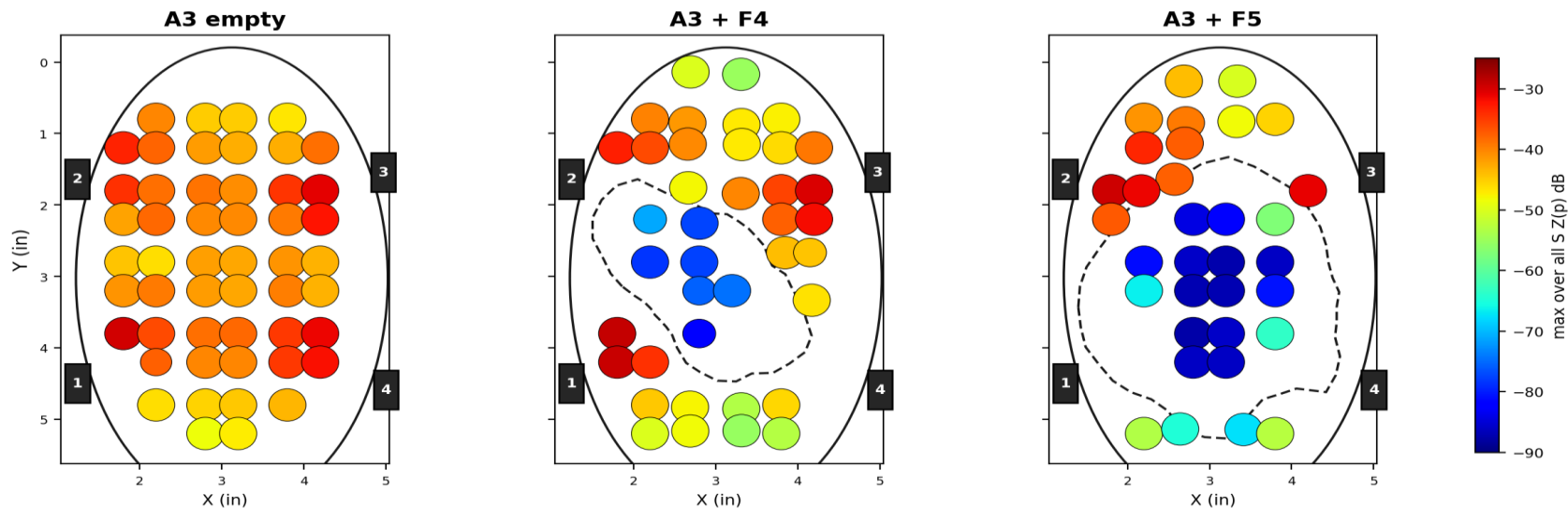
Mean ± std over LOSO folds. Bold = better method.

With the full antenna array, both methods are near-perfect on every phantom. The CNN stays at 100% on F5; the MLP gives up ~5 points there under pipeline v2.

Error Locations: all 4 antennas

Dot color = how strongly the tumor physically changes the signal (detectable difference). **Dot size** = CNN LOSO accuracy at that position. Dashed outline = glandular insert.

Detectable change (color, max over all S) vs CNN LOSO accuracy (dot size) -- all 4 antennas



Detectable change still dips inside the glandular insert (blue), but with 16 complementary views the classifier compensates: accuracy dots stay large almost everywhere. MLP versions look qualitatively identical.

All 4 antennas: reflection only (S11, S22, S33, S44)

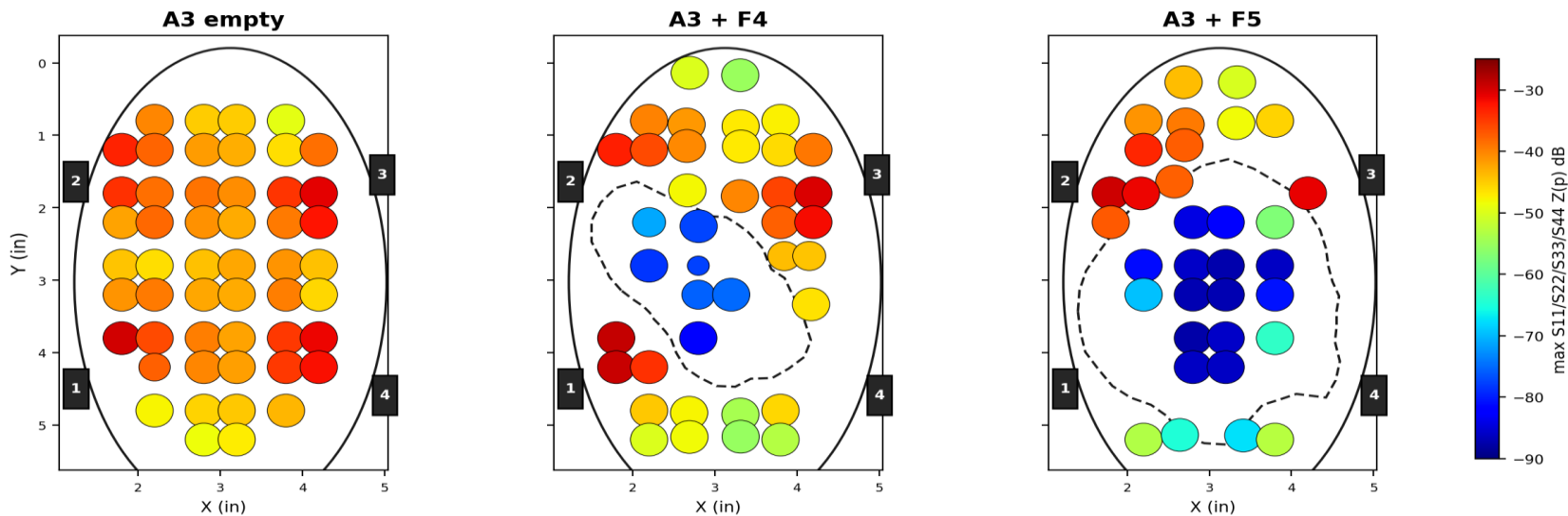
Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	99.3 ± 1.1	99.3 ± 0.9	+0.0
A3 + F4	39-way	95.5 ± 3.9	93.6 ± 3.9	+1.9
A3 + F5	33-way	100.0 ± 0.0	91.9 ± 3.8	+8.1

No through-phantom transmission paths; each antenna only listens to its own echo.

Dropping transmission barely touches the CNN (F5 stays at 100%), while the MLP gives up 3-5 points on the glandular phantoms. Four reflection coefficients are enough for a CNN.

Error Locations: all 4 antennas, reflection only

Detectable change (color, max S11/S22/S33/S44) vs CNN LOSO accuracy (dot size) -- all 4 antennas, reflection only



Reflection-only detectable change (max of S11, S22, S33, S44). The insert still carves a low-signal pocket, but CNN accuracy dots stay large.

2 antennas (ports 1 & 3): full S-parameters

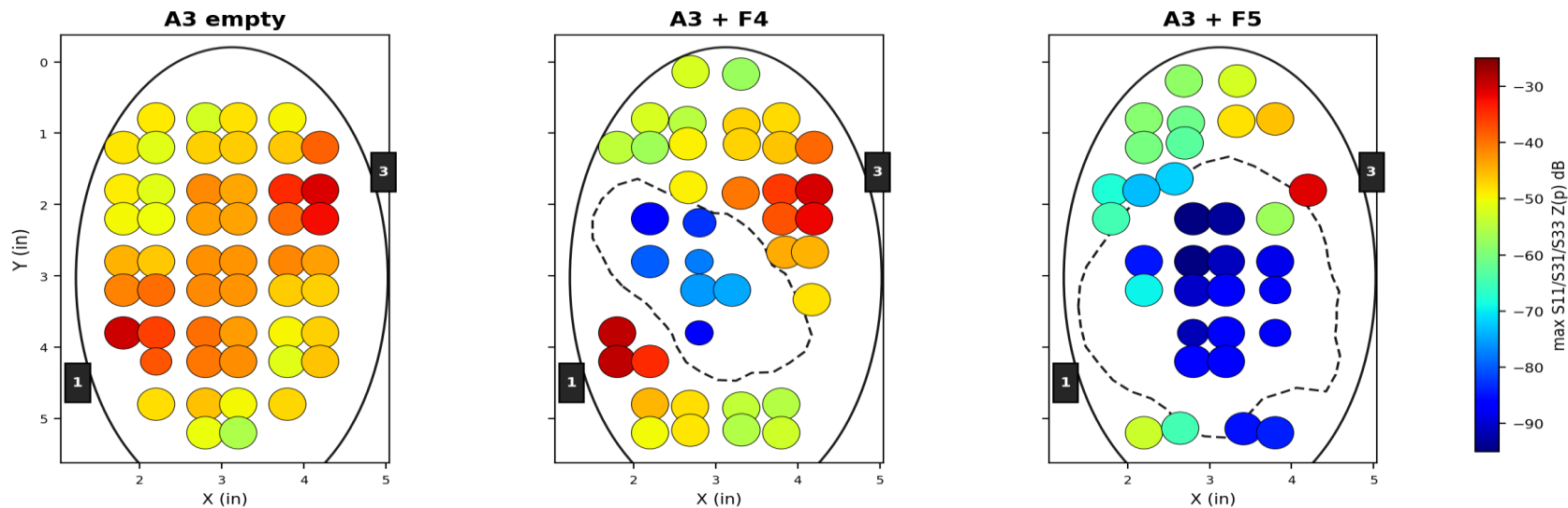
Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	99.3 ± 1.1	99.3 ± 0.9	+0.0
A3 + F4	39-way	96.8 ± 2.5	96.8 ± 2.1	+0.0
A3 + F5	33-way	97.0 ± 5.3	84.9 ± 6.6	+12.1

Antennas 1 & 3 sit on opposite sides of the phantom (most informative pair).

Halving the antennas separates the methods on F5: the CNN holds 97% while the MLP drops to 85%. On F4 the two are tied under pipeline v2.

Error Locations: 2 antennas (1 & 3)

Detectable change (color, max S11/S31/S33) vs CNN LOSO accuracy (dot size) -- antennas 1 & 3



With two antennas a low-signal pocket appears inside the glandular insert, and accuracy dots shrink with it.

2 antennas (1 & 3): reflection only (S11, S33)

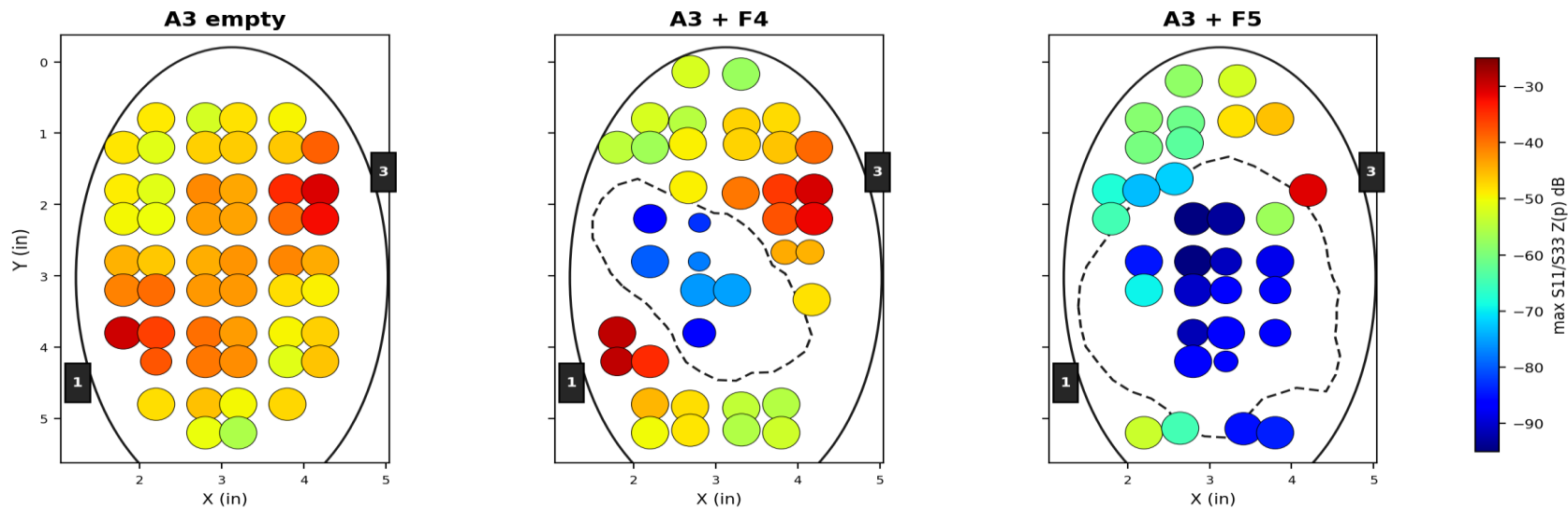
Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	99.3 ± 1.1	99.3 ± 0.9	+0.0
A3 + F4	39-way	91.7 ± 2.5	92.3 ± 3.6	−0.6
A3 + F5	33-way	92.9 ± 4.6	80.8 ± 7.1	+12.1

The toughest 2-antenna condition: two reflection coefficients and nothing else.

The CNN holds 92-99% while the MLP falls to 81% on F5. F4 remains the single config in the study where the MLP edges the CNN (92.3 vs 91.7).

Error Locations: 2 antennas (1 & 3), reflection only

Detectable change (color, max S11/S33) vs CNN LOSO accuracy (dot size) -- antennas 1 & 3, reflection only



With only S11 and S33, the low-signal pocket deepens and CNN errors start to appear inside the insert.

1 antenna: S11 only

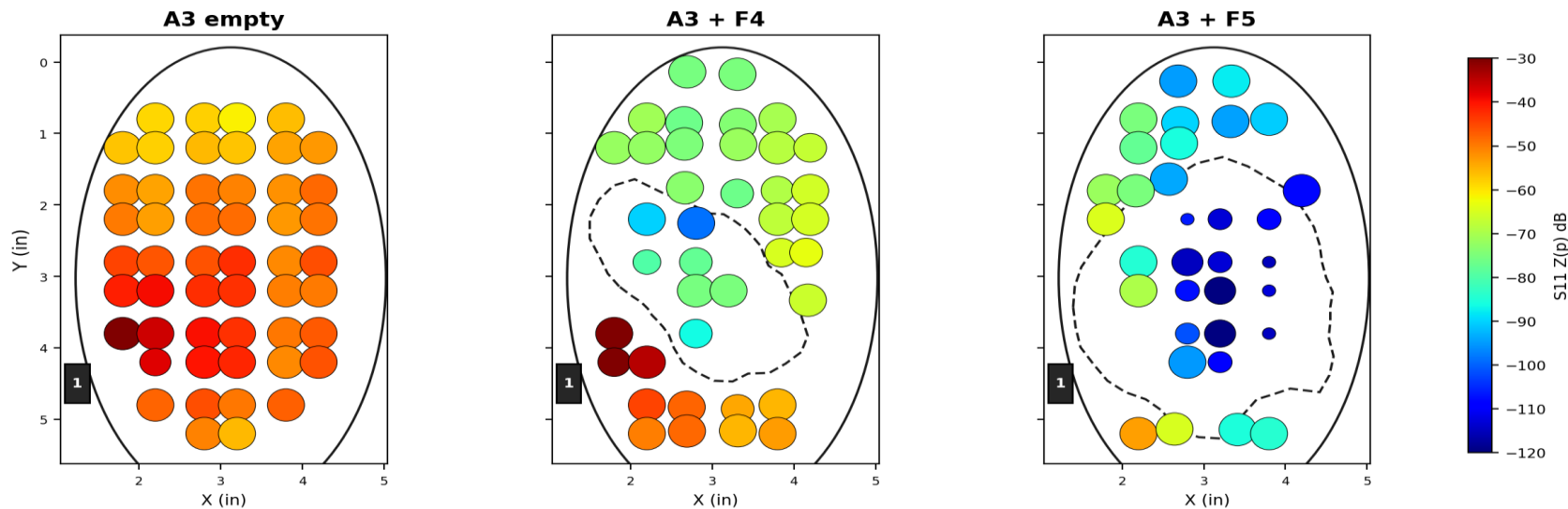
Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	99.3 ± 1.1	99.3 ± 0.9	+0.0
A3 + F4	39-way	92.9 ± 7.4	90.4 ± 8.0	+2.5
A3 + F5	33-way	72.7 ± 6.1	64.7 ± 5.7	+8.0

A single reflection coefficient: the minimal possible hardware.

The extreme case: one antenna, one S-parameter. The CNN holds 73–99%; the MLP falls to 65% on F5. The gap grows as the task gets harder, and the F5 errors cluster inside the glandular insert (next slide).

Error Locations: 1 antenna (S11)

Detectable change (color, S11) vs CNN LOSO accuracy (dot size) -- antenna 1 (S11)



Single antenna (S11): small blue dots cluster inside the insert. Where physics buries the tumor, the classifier fails.

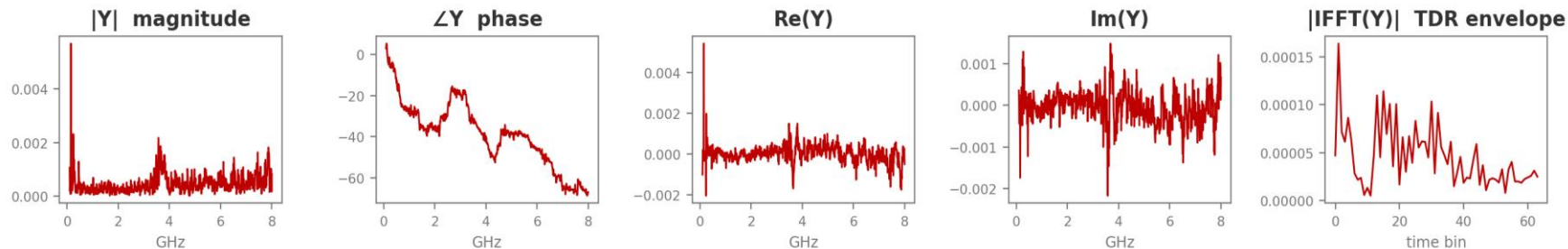
PART 4

Physics-informed features

Squeezing more out of the same measurement: no new hardware, no new data collection

One complex measurement, five physical views

Each calibrated S-parameter $Y(f)$ is one complex spectrum. From it we derive, at zero measurement cost: magnitude & phase (resonance behaviour), real & imaginary parts (full complex information), and the inverse-FFT envelope, a time-domain reflectometry (TDR) view where the tumor appears as an echo at its round-trip delay.



Real example: A3+F5 phantom, S11, tumor at grid position R3C4 (calibrated response Y)

The 'physics' input feeds all five views to both models: the CNN as extra image rows, the MLP as extra vector features. Identical information, matched again.

Same tests, physics inputs

All-4-antenna physics: both models at ceiling, 97–100% (F5 reaches 100 / 100). Below: the configs where it matters. Reflection-only: next slide.

2 antennas (1 & 3), physics input

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	99.3 ± 1.1	99.3 ± 0.9	+0.0
A3 + F4	39-way	98.7 ± 2.6	97.4 ± 2.6	+1.3
A3 + F5	33-way	99.0 ± 1.8	89.9 ± 7.6	+9.1

1 antenna (S11), physics input

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	99.3 ± 1.1	98.7 ± 1.9	+0.6
A3 + F4	39-way	95.5 ± 3.2	93.6 ± 4.6	+1.9
A3 + F5	33-way	76.8 ± 3.5	66.7 ± 8.9	+10.1

Reflection only + physics features

All 4 antennas, reflection only (S11, S22, S33, S44), physics input

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	99.3 ± 1.1	99.3 ± 0.9	+0.0
A3 + F4	39-way	96.2 ± 3.3	94.2 ± 2.8	+2.0
A3 + F5	33-way	100.0 ± 0.0	88.9 ± 5.2	+11.1

2 antennas (1 & 3), reflection only (S11, S33), physics input

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	99.3 ± 1.1	98.0 ± 1.6	+1.3
A3 + F4	39-way	94.2 ± 2.5	91.7 ± 4.6	+2.5
A3 + F5	33-way	98.0 ± 3.5	81.8 ± 4.3	+16.2

With physics features, reflection-only costs the CNN almost nothing (F5: 100% with four antennas, 98% with two); the MLP pays 8-11 points. A reflection-only system is practical with a CNN.

What the derived features buy, and what TDR alone can't

Physics vs raw (percentage points gained)

- F5, 2 antennas: CNN +2.0 (97→99) · MLP +5.1 (85→90)
- F5, 1 antenna: CNN +4.0 (73→77) · MLP +2.0 (65→67)
- F5, all 4: MLP +4.0 (95→99); CNN already at 100
- Easy setups (empty, F4 full-array): no change, already at ceiling.
- **Physics features help most exactly where raw inputs struggle.**

TDR-only ablation (selected)

- F4, 1 antenna: CNN 55 / MLP 64 (vs 93 / 90 raw): a big loss for both.
- TDR is where v2 hurts: removing the mean subtraction leaves coherent session clutter in the envelope (CNN F5-pair tdr fell 79→63).

Additional finding: hierarchical (quadrant → position) classification

- Two-stage 'find the corner first' scheme tested on both models.
- It never helps, and hurts the hardest cases: CNN -6 on F5, MLP -6 on F4-single.
- **Wrong quadrant = unrecoverable. Single-stage wins under LOSO.**

What removing the mean subtraction changed (v1 vs v2)

MLP: clear gains

- F5 single raw: 58.6 $\rightarrow$ 64.7 (+6.1)
- F5 refl-pair raw: 74.8 $\rightarrow$ 80.8 (+6.1)
- F4 pair physics: 92.3 $\rightarrow$ 97.4 (+5.1)
- F5 pair physics: 85.9 $\rightarrow$ 89.9 (+4.0)
- F4 pair raw: 93.6 $\rightarrow$ 96.8 (+3.2)
- Empty phantom: unchanged (ceiling).

CNN: mixed, mostly neutral

- Most raw / physics configs: within ± 1 point of v1.
- Single-antenna dips: F4 raw 96.8 $\rightarrow$ 92.9, F5 raw 75.8 $\rightarrow$ 72.7, F5 physics 79.8 $\rightarrow$ 76.8.
- TDR-only suffers badly: F5 pair 78.8 $\rightarrow$ 62.6; F4 single 84.0 $\rightarrow$ 55.1 (a fold collapse returns).

Verdict: the mean subtraction was redundant for frequency-domain inputs once the per-session z-score exists (the MLP gains by dropping it), but TDR inputs genuinely need it: coherent session clutter must be removed BEFORE the envelope is computed. Best practice going forward: skip it for raw/physics, keep it for TDR.

CONCLUSIONS

What we learned

CNN $\geq$ MLP, everywhere

On matched inputs the CNN ties easy cases and wins hard ones by up to 16 points, at $\sim 10\times$ the training cost.

The gap grows with difficulty

Fewer antennas + bigger glandular insert = bigger CNN advantage.

Preprocessing drives robustness

Per-session z-score eliminated the CNN's cross-session failures; the v1 mean-sub step proved unnecessary.

Physics features add accuracy at no cost

Derived views (+TDR) add up to 4 points on the hardest configurations.

Errors follow the physics

Misses cluster inside the glandular insert, where detectable signal change collapses.

Fewer antennas is viable with a CNN

2 antennas: 97–99% on every phantom. Even 1 antenna: 73–99%.